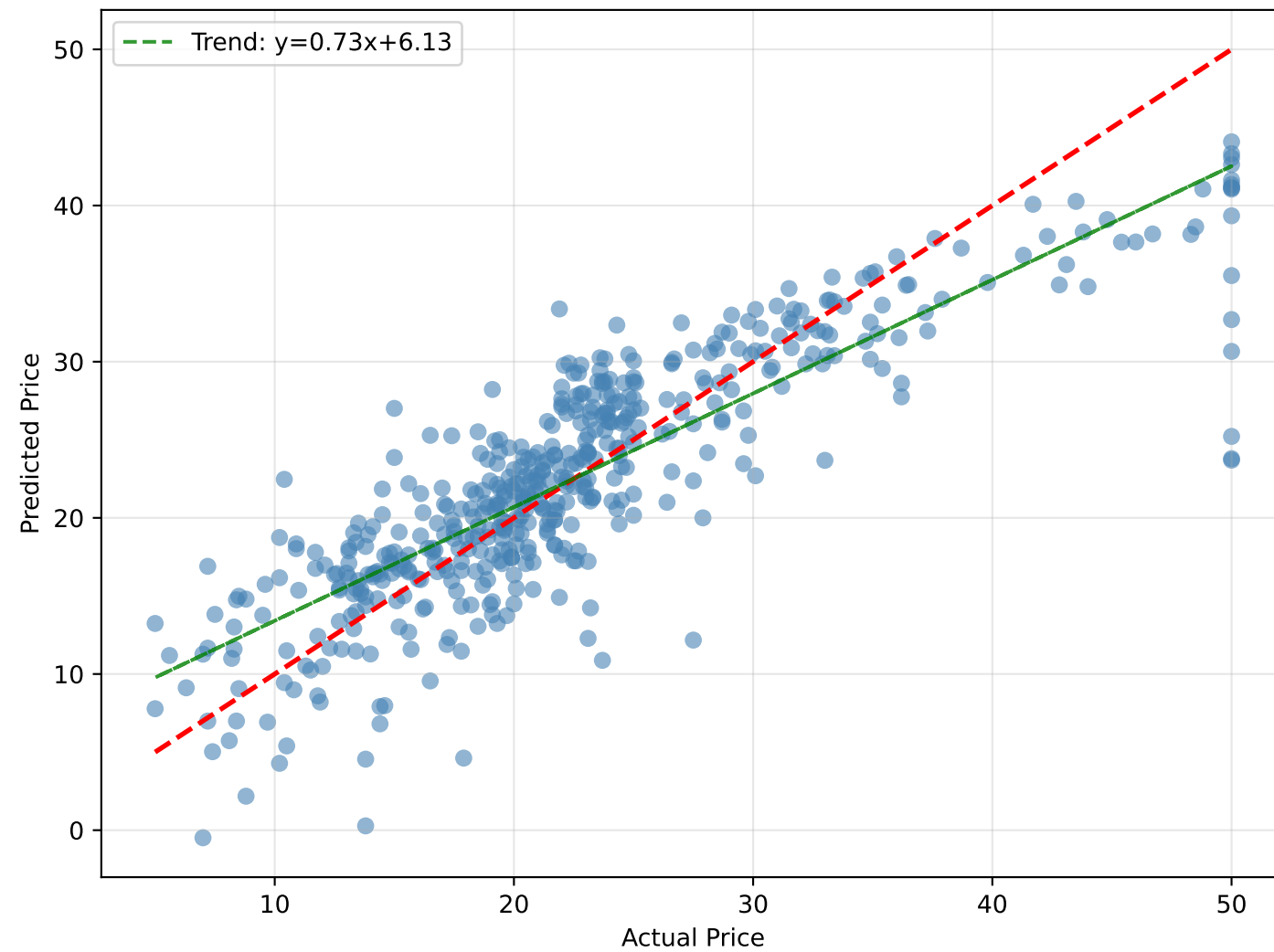
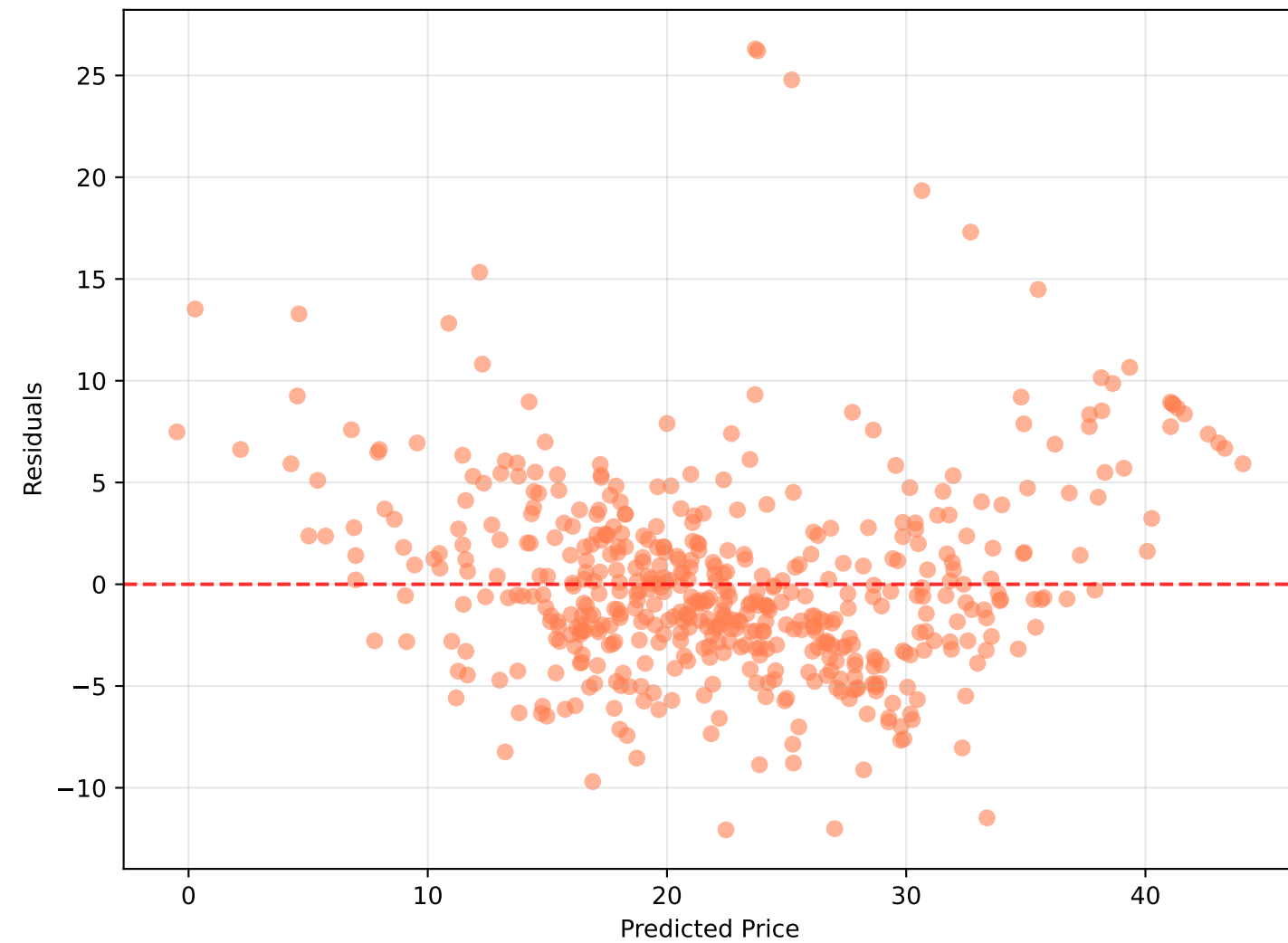


Overall Regression Analysis: Price Behavior

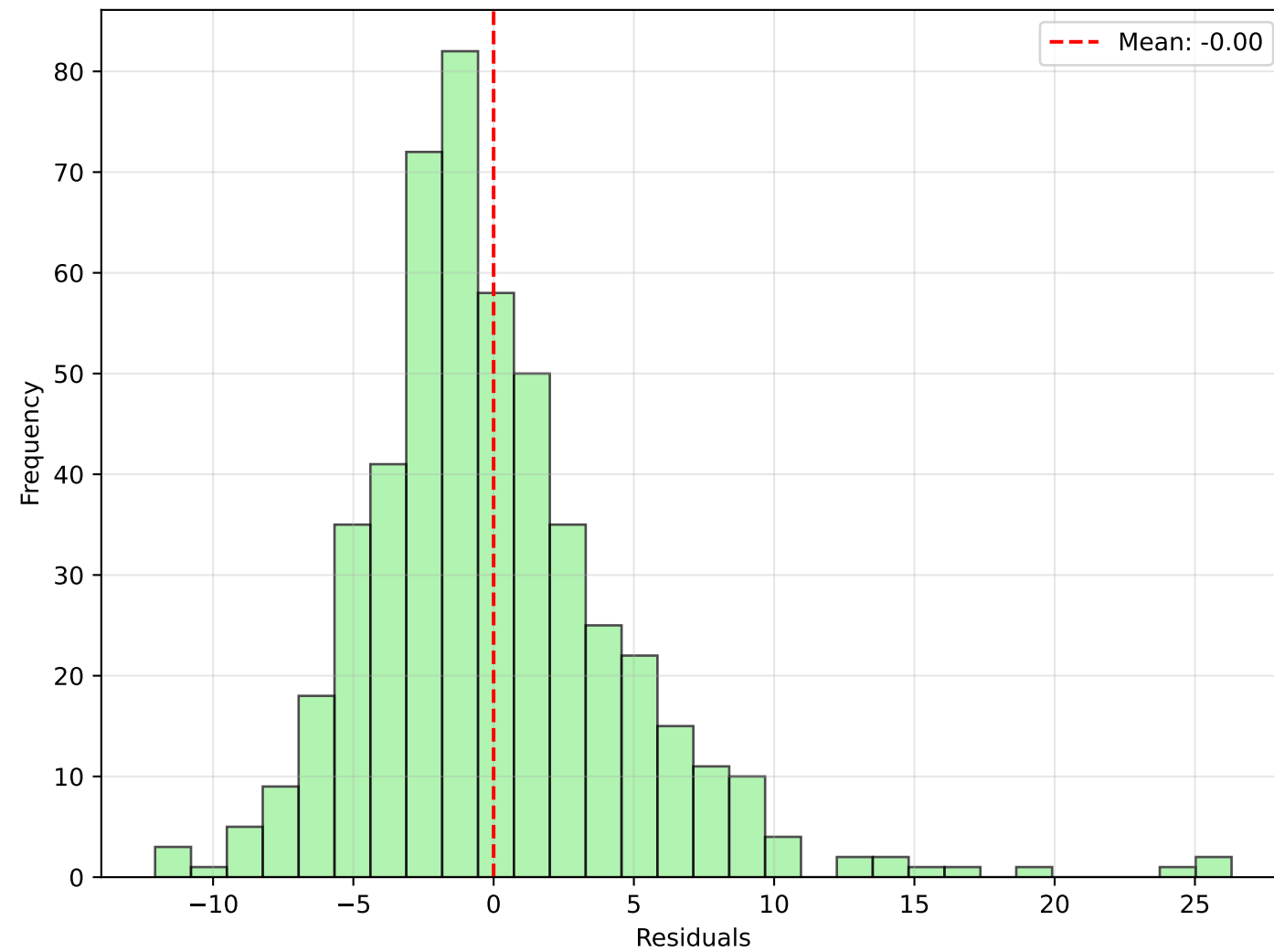
Predicted vs Actual ($R^2 = 0.728$)



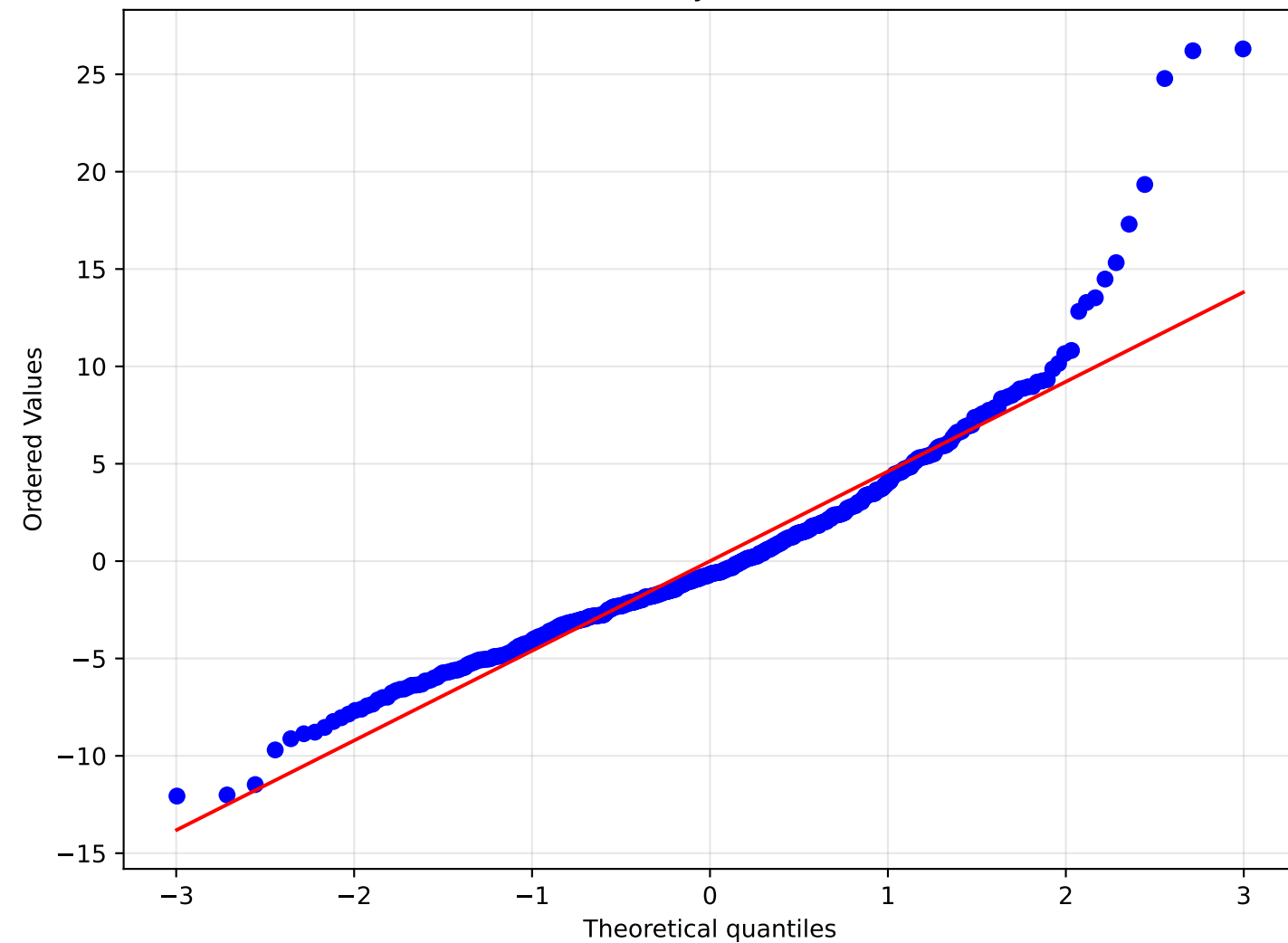
Residuals vs Predicted



Distribution of Residuals

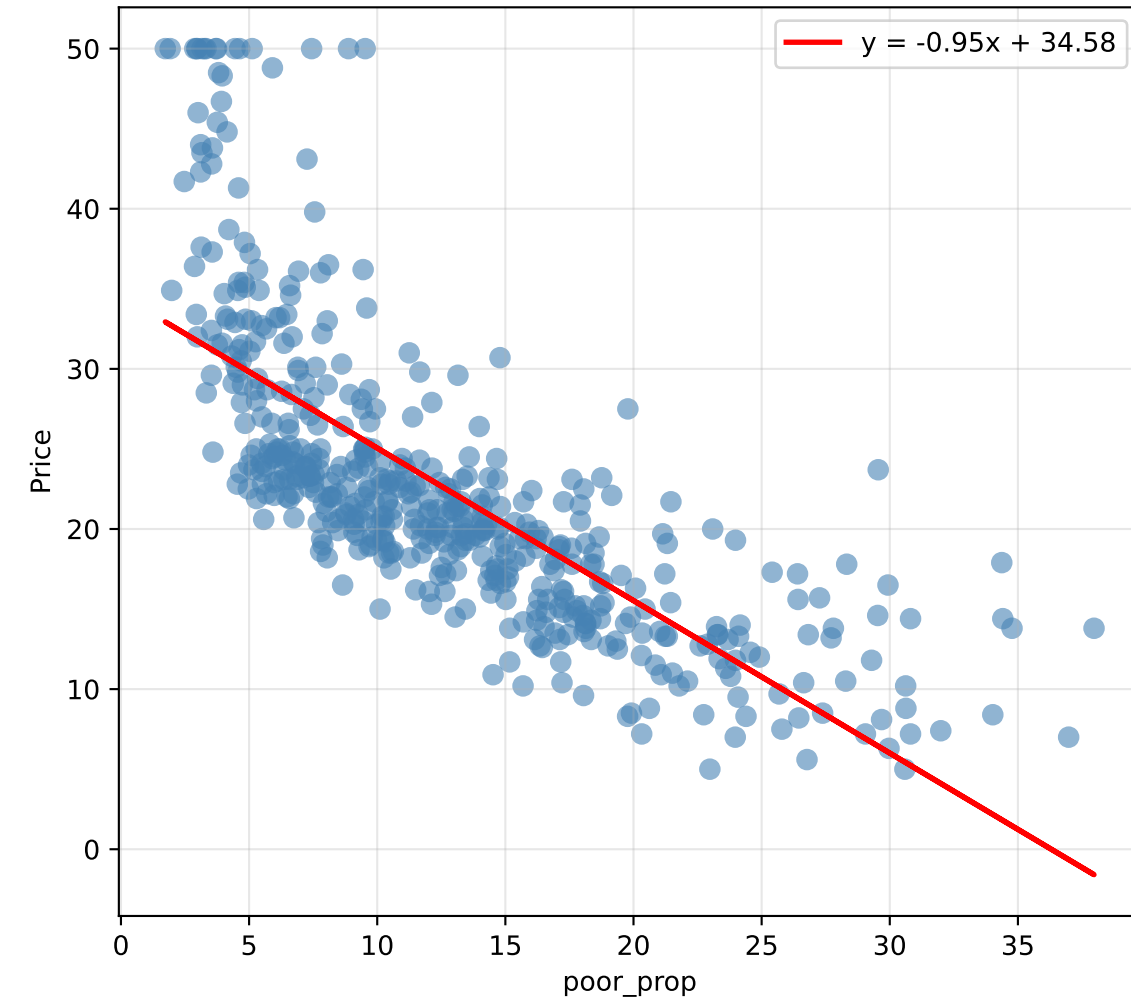


Q-Q Plot: Normality Check of Residuals

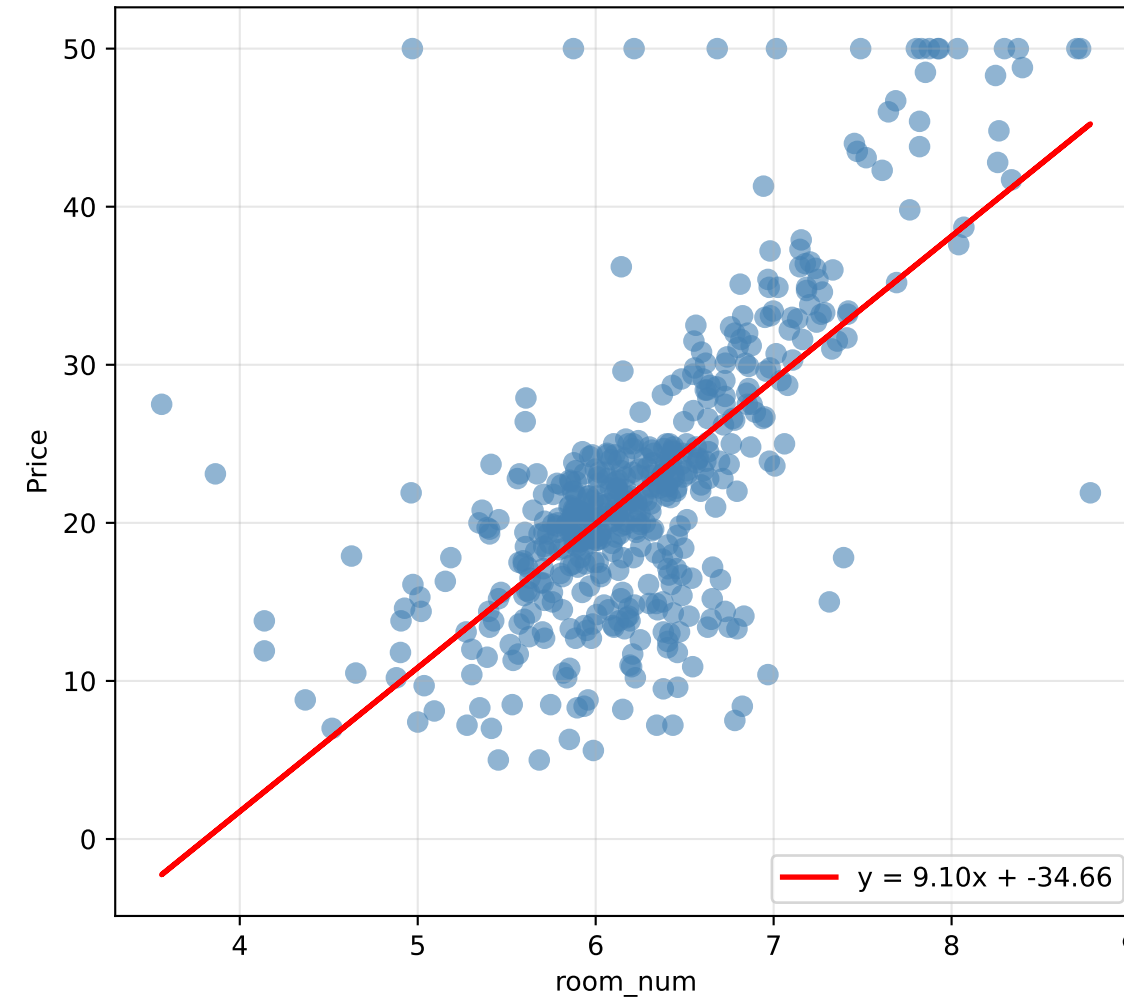


Individual Feature Regression: Relationship with Price

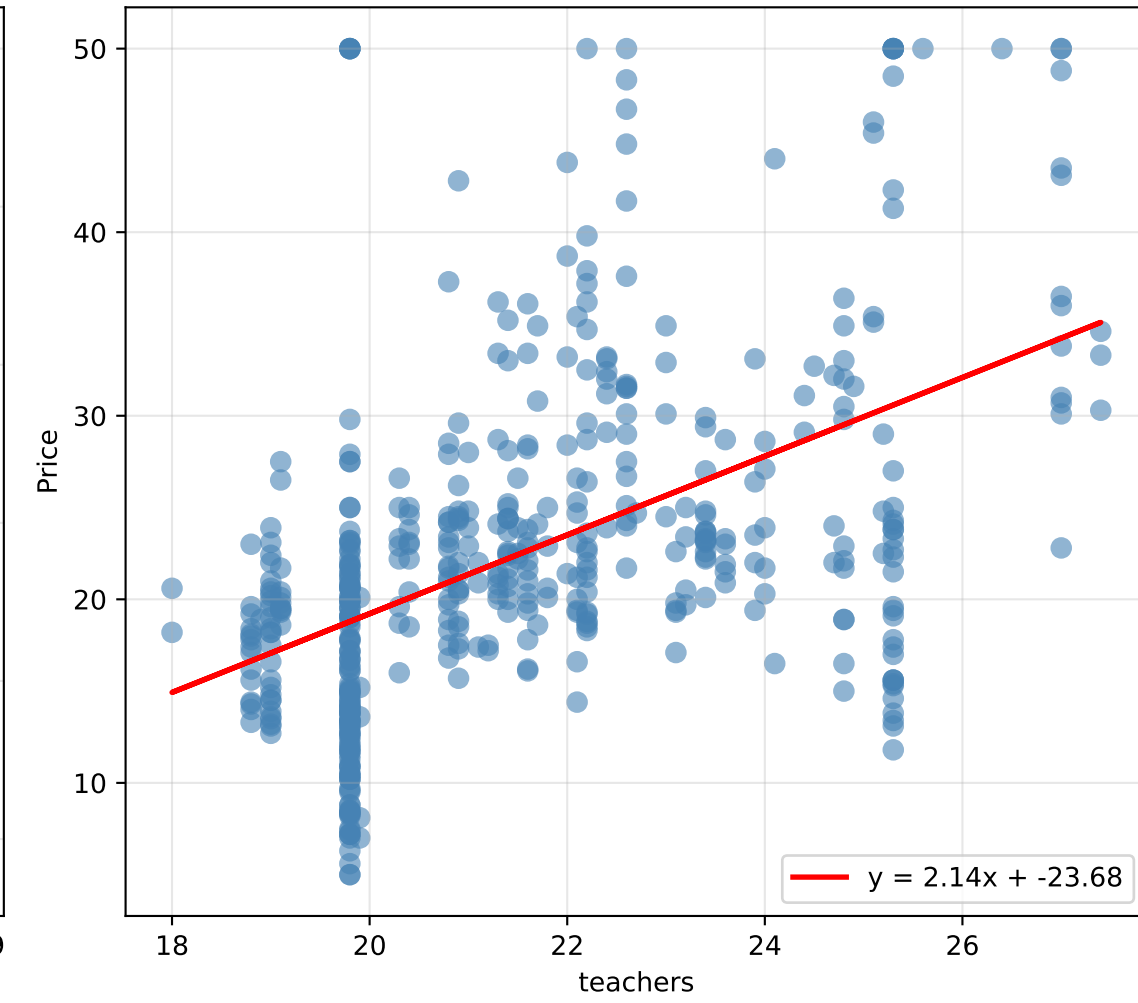
poor_prop vs Price
($R = -0.741$, $p = 3.732e-89$)



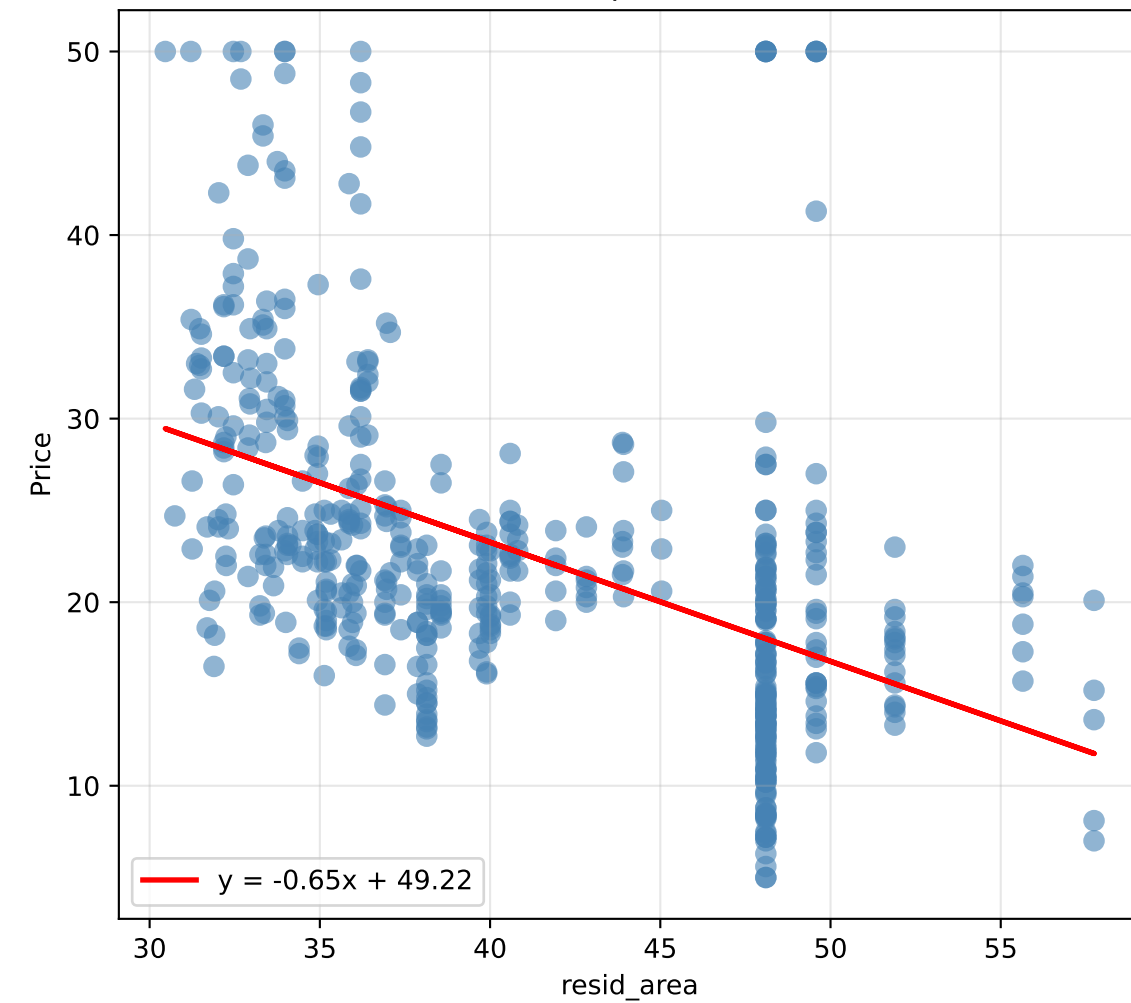
room_num vs Price
($R = 0.696$, $p = 1.307e-74$)



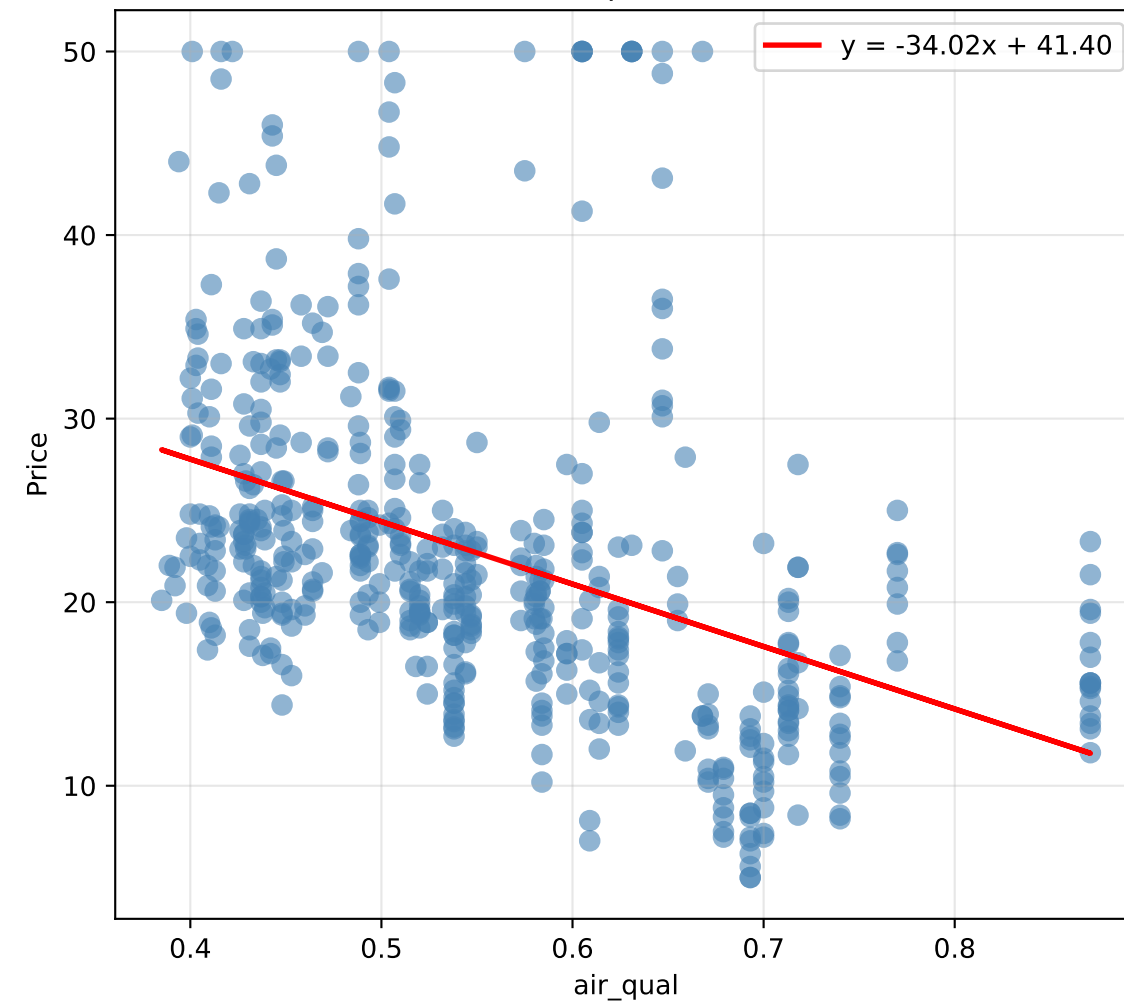
teachers vs Price
($R = 0.506$, $p = 3.363e-34$)



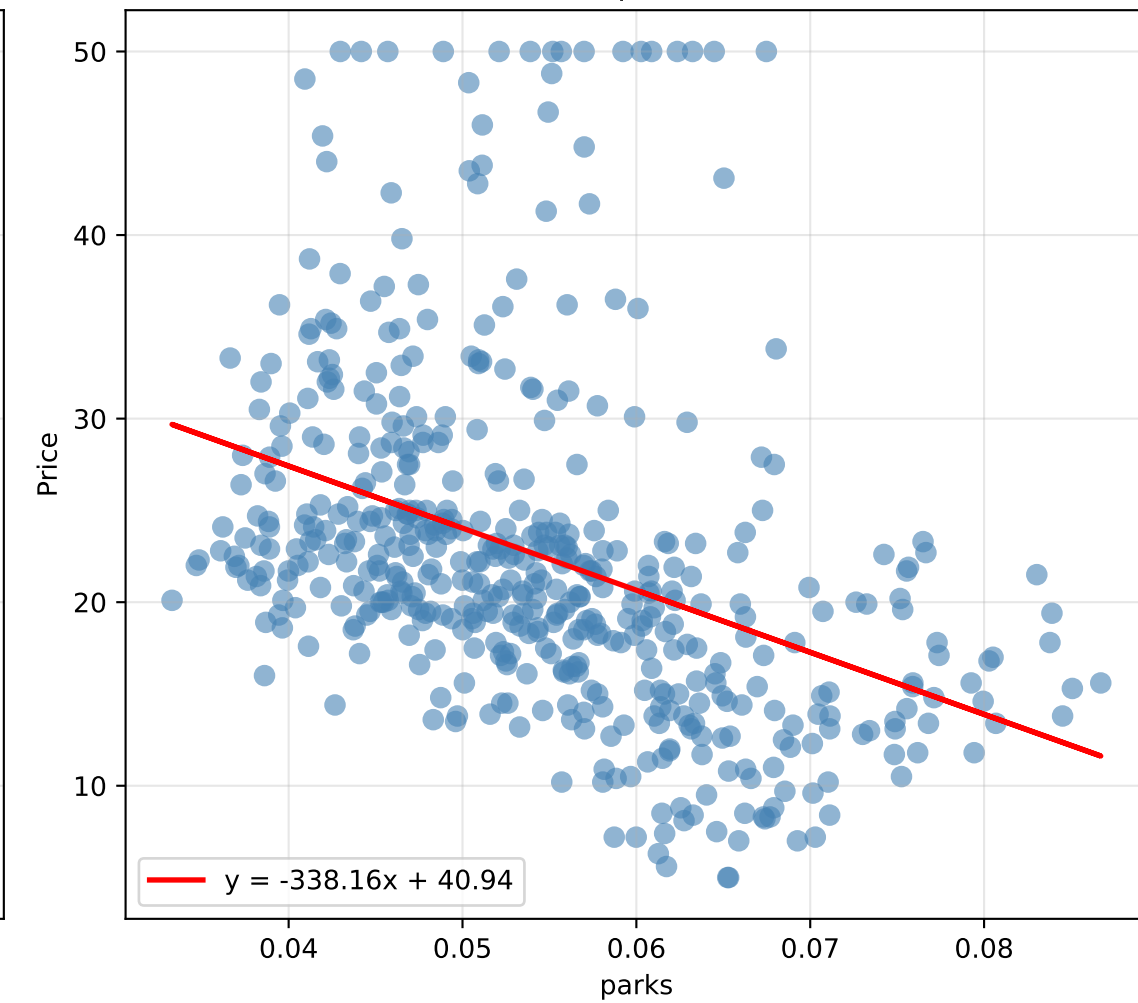
resid_area vs Price
($R = -0.485$, $p = 3.522e-31$)



air_qual vs Price
($R = -0.429$, $p = 4.168e-24$)

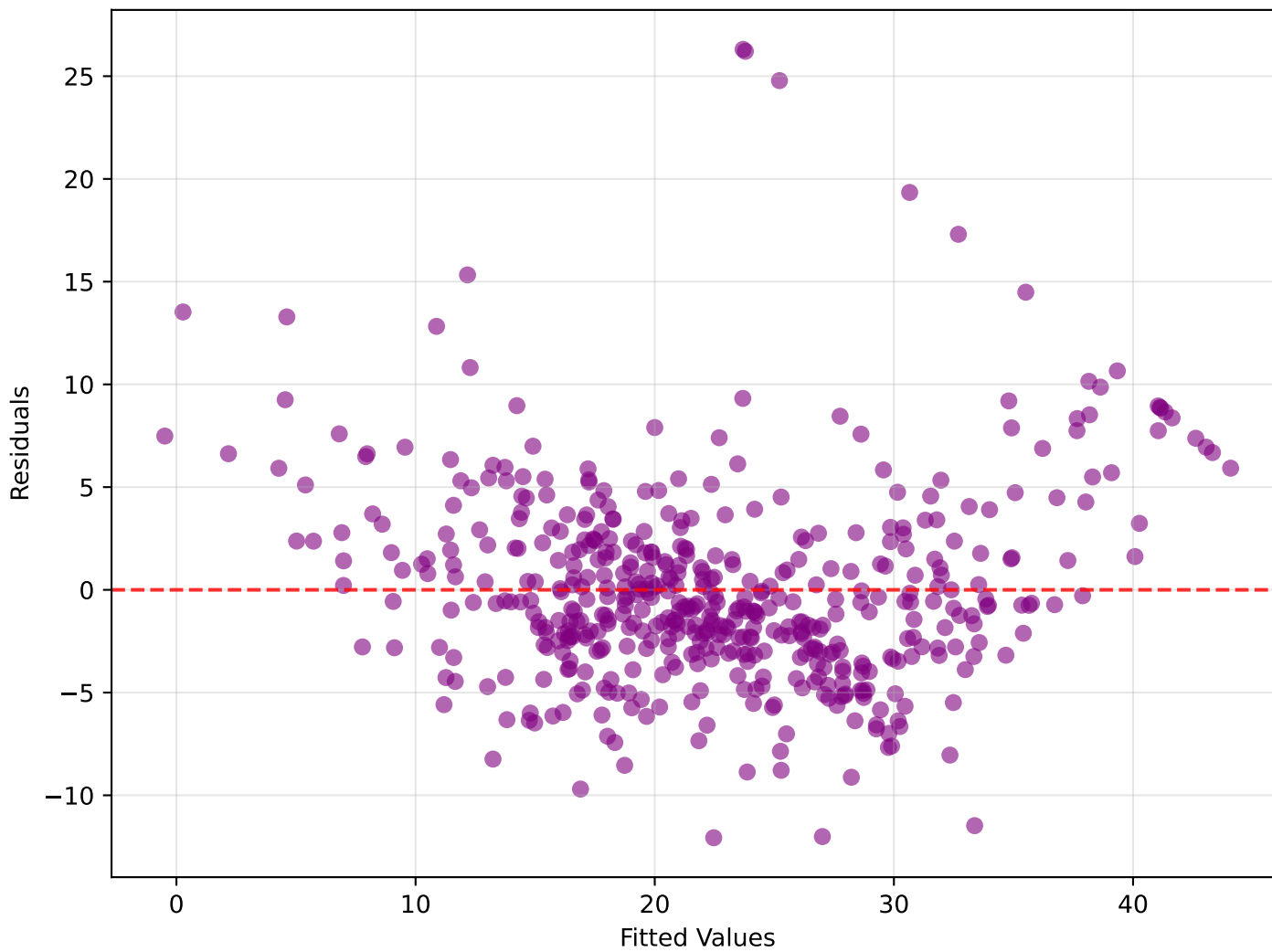


parks vs Price
($R = -0.392$, $p = 5.458e-20$)

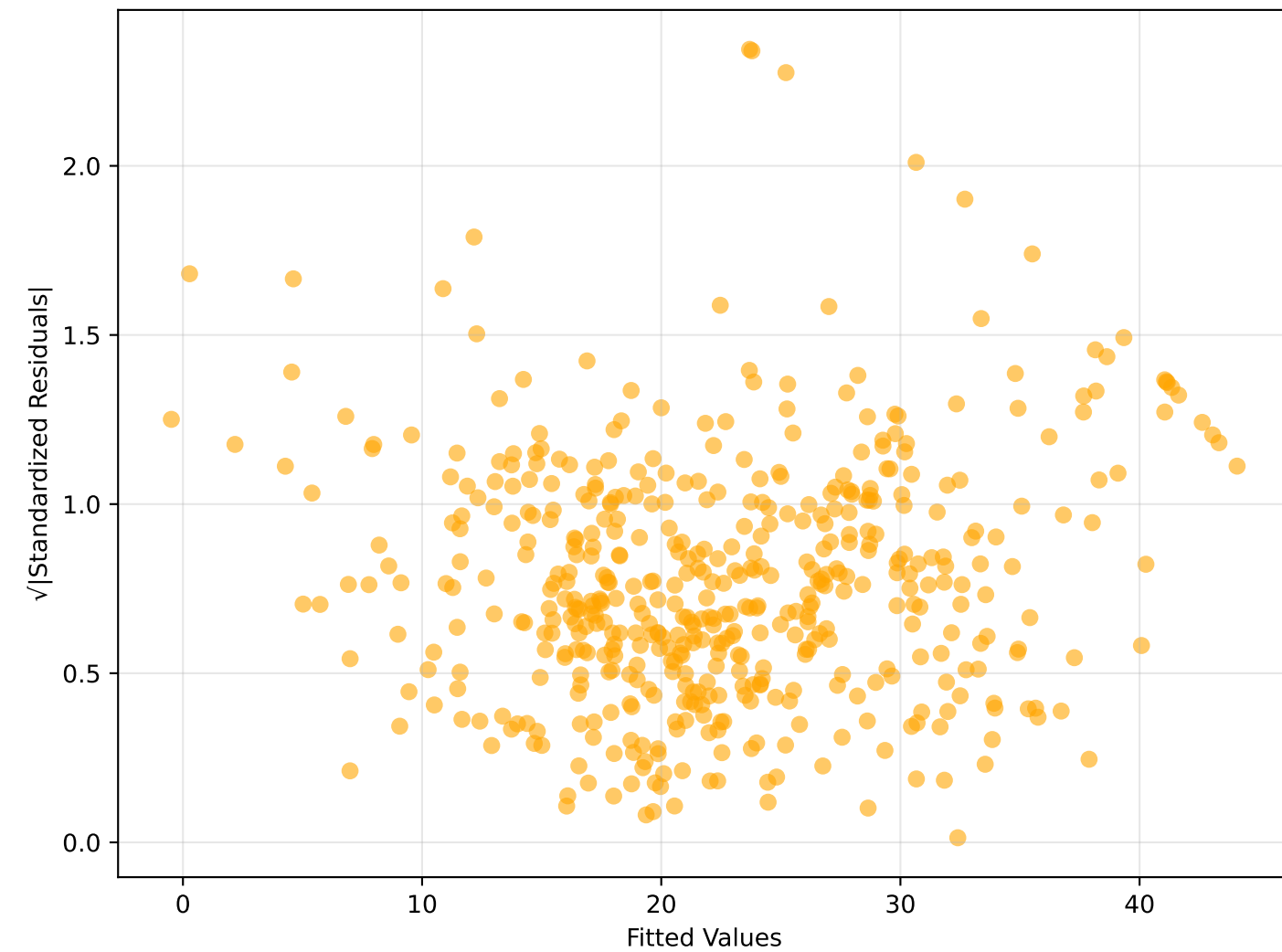


Comprehensive Residual Analysis

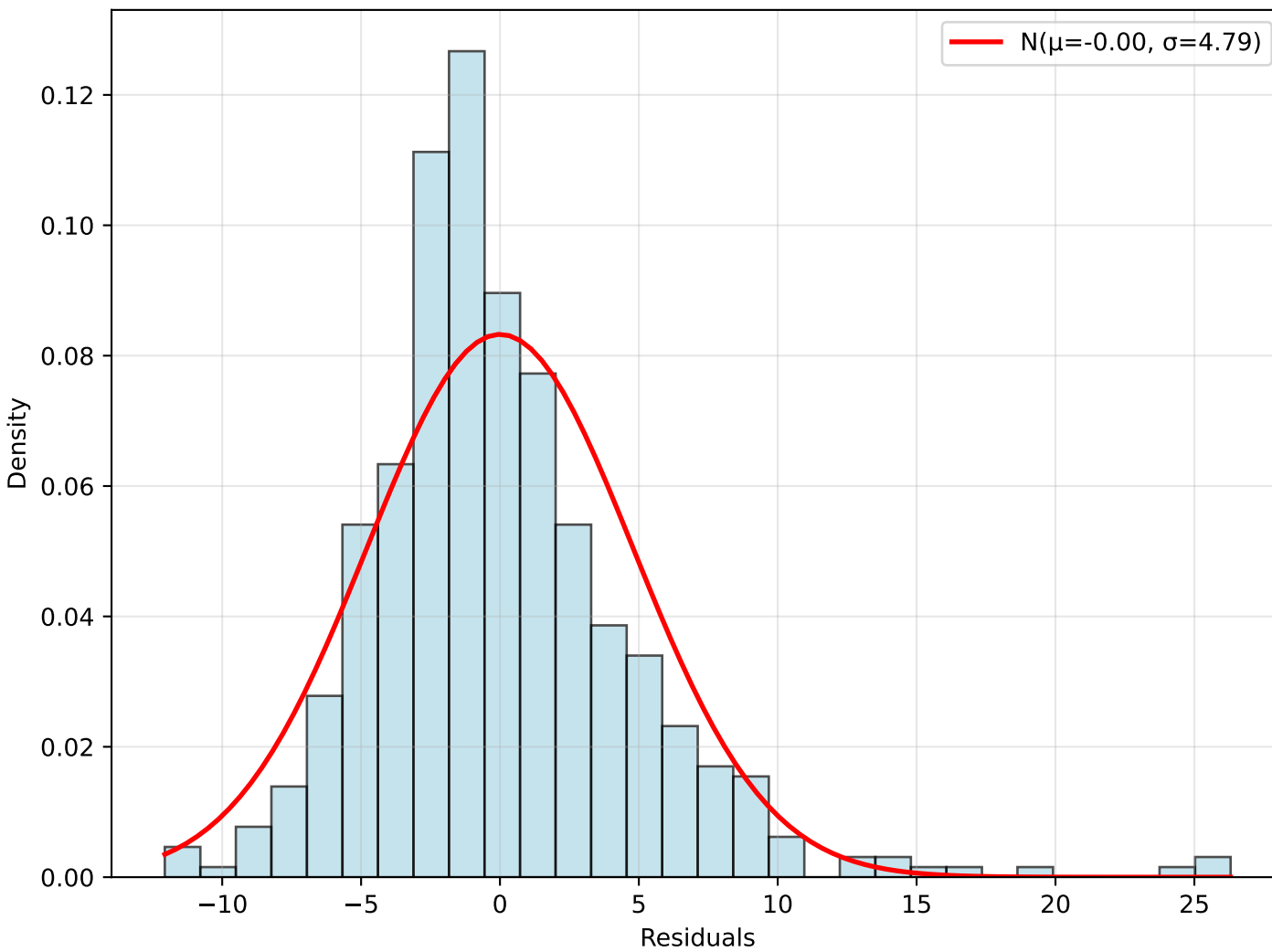
Residuals vs Fitted Values



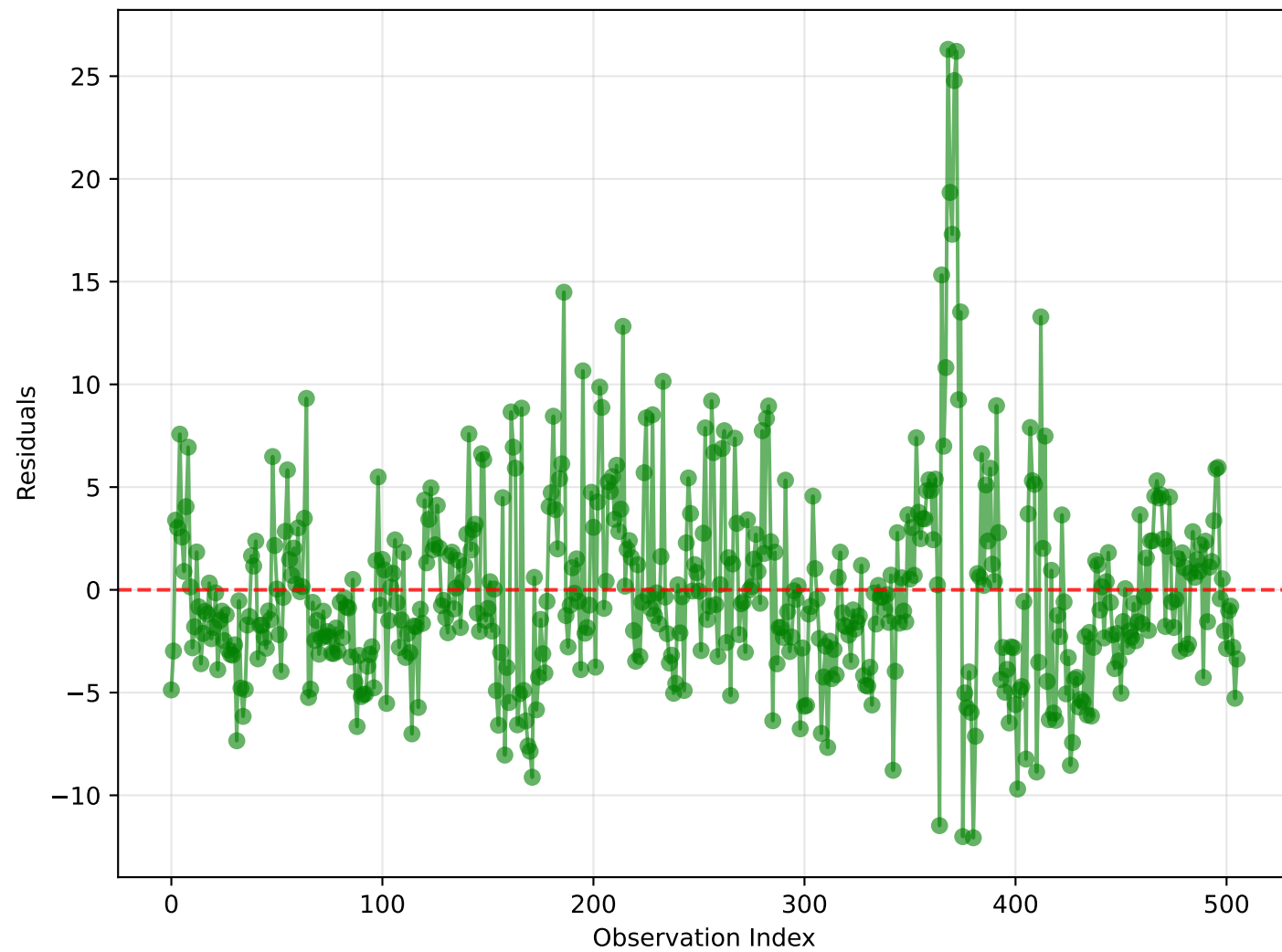
Scale-Location Plot



Residual Distribution vs Normal

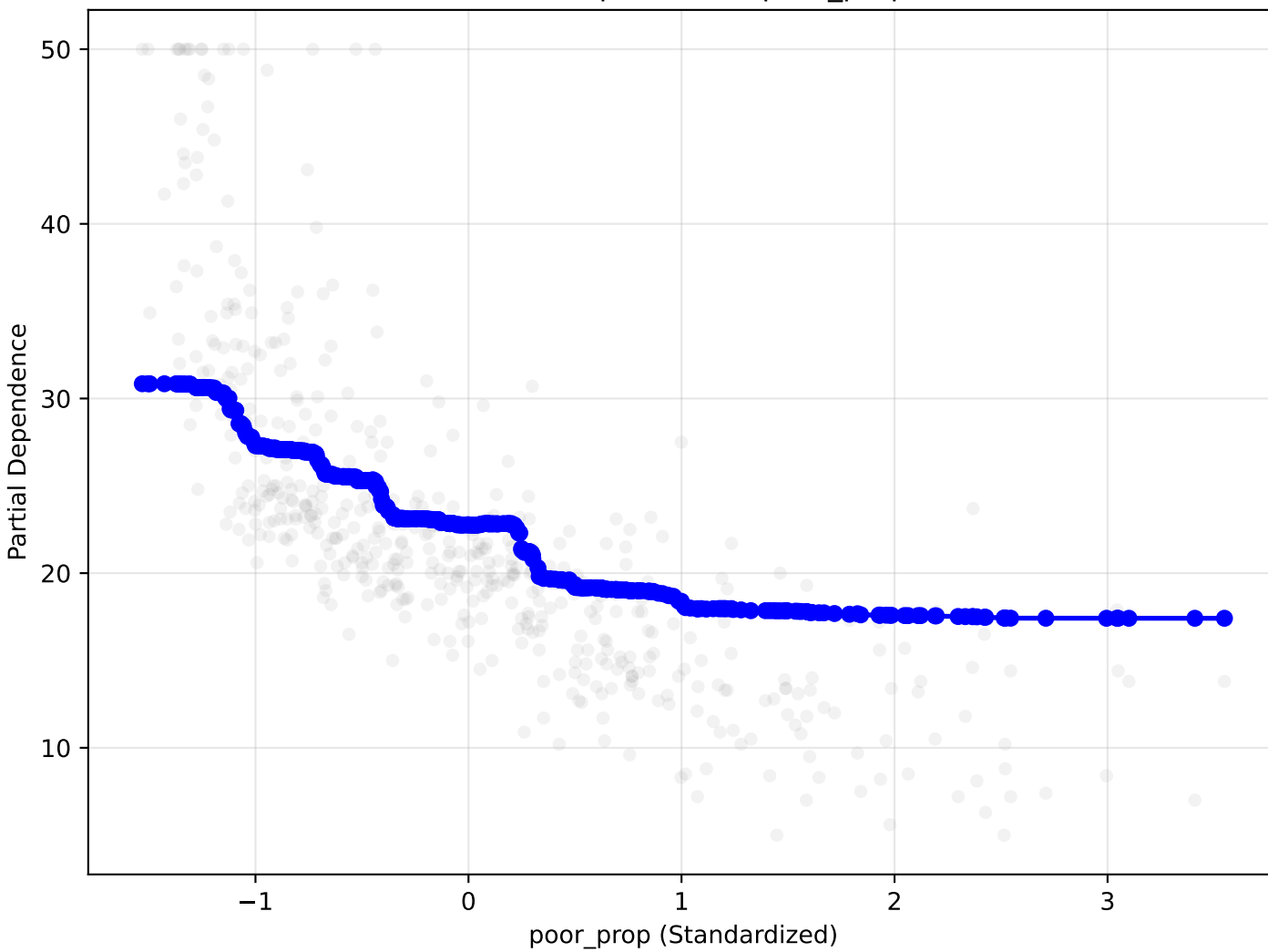


Residuals Order Plot

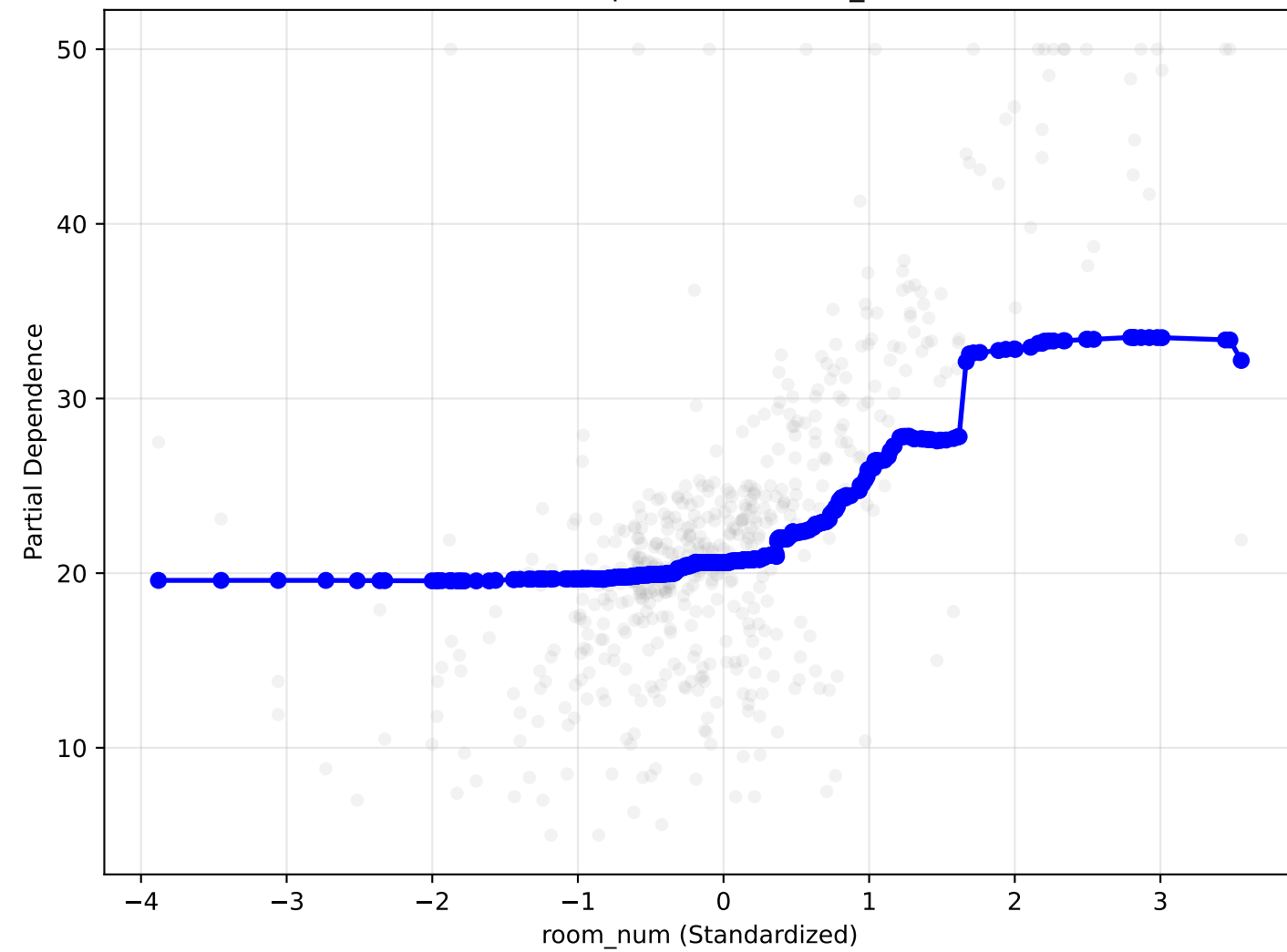


Partial Dependence Plots: Marginal Effect on Price

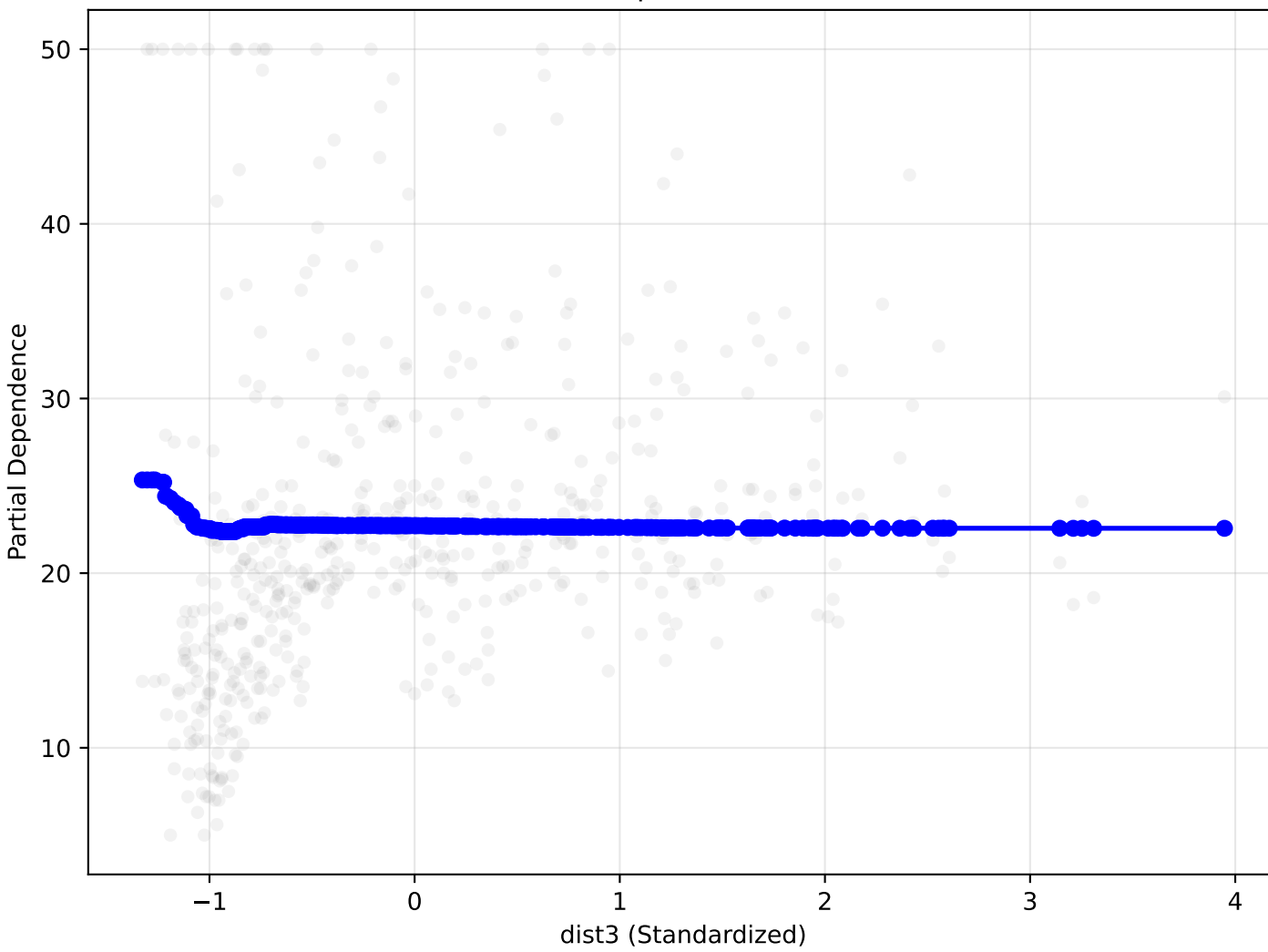
Partial Dependence: poor_prop



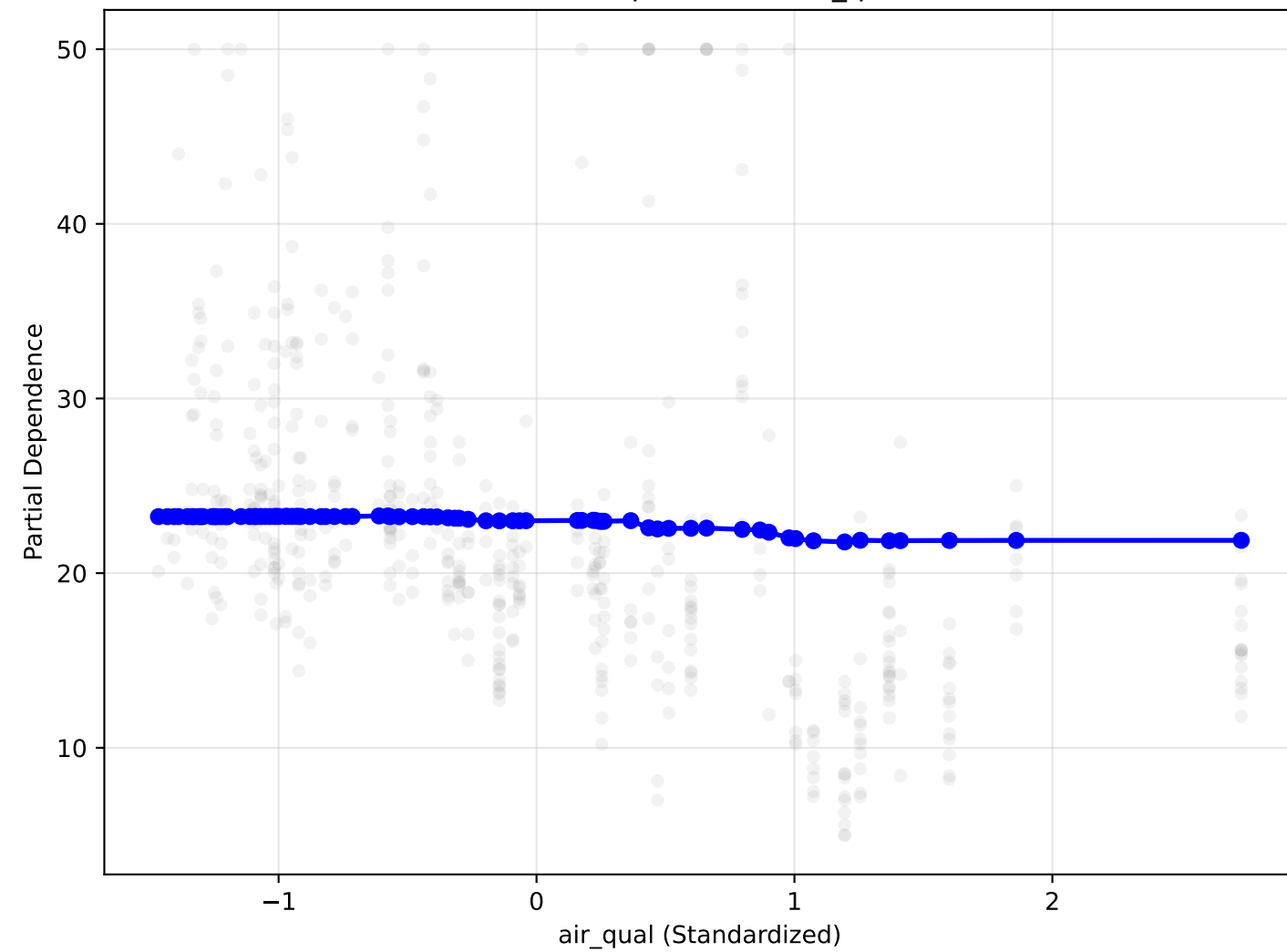
Partial Dependence: room_num



Partial Dependence: dist3

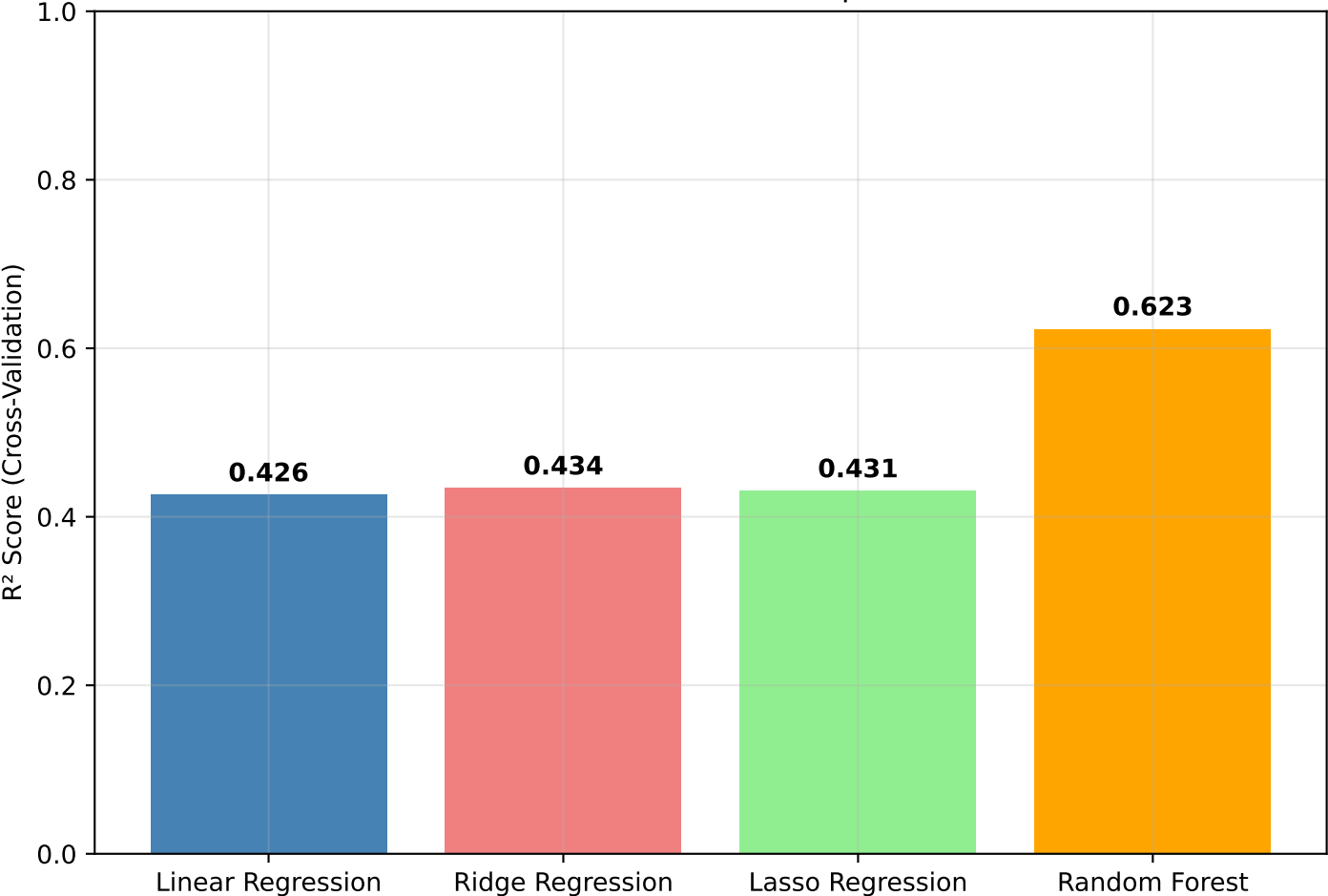


Partial Dependence: air_qual

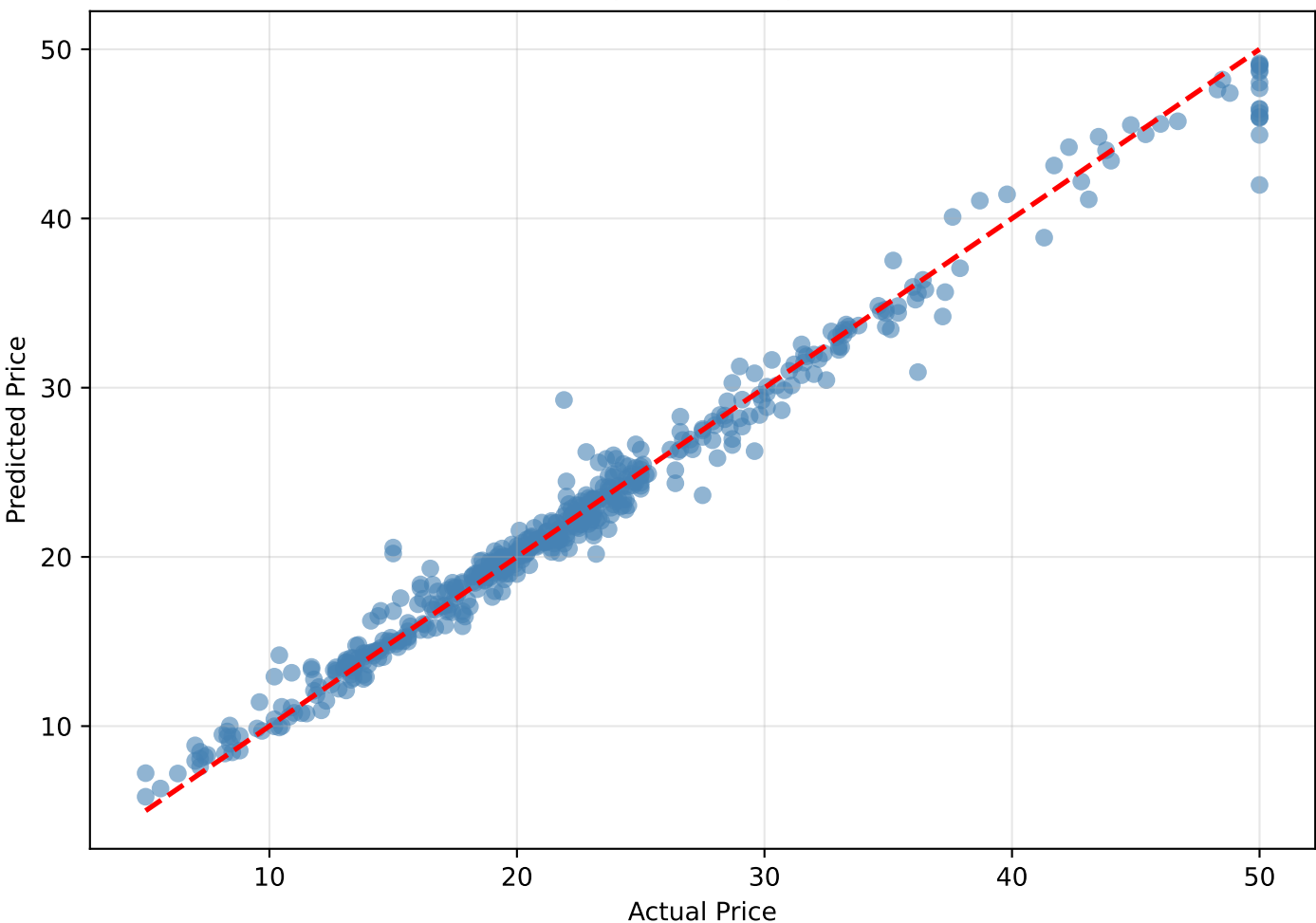


Regression Model Comparison

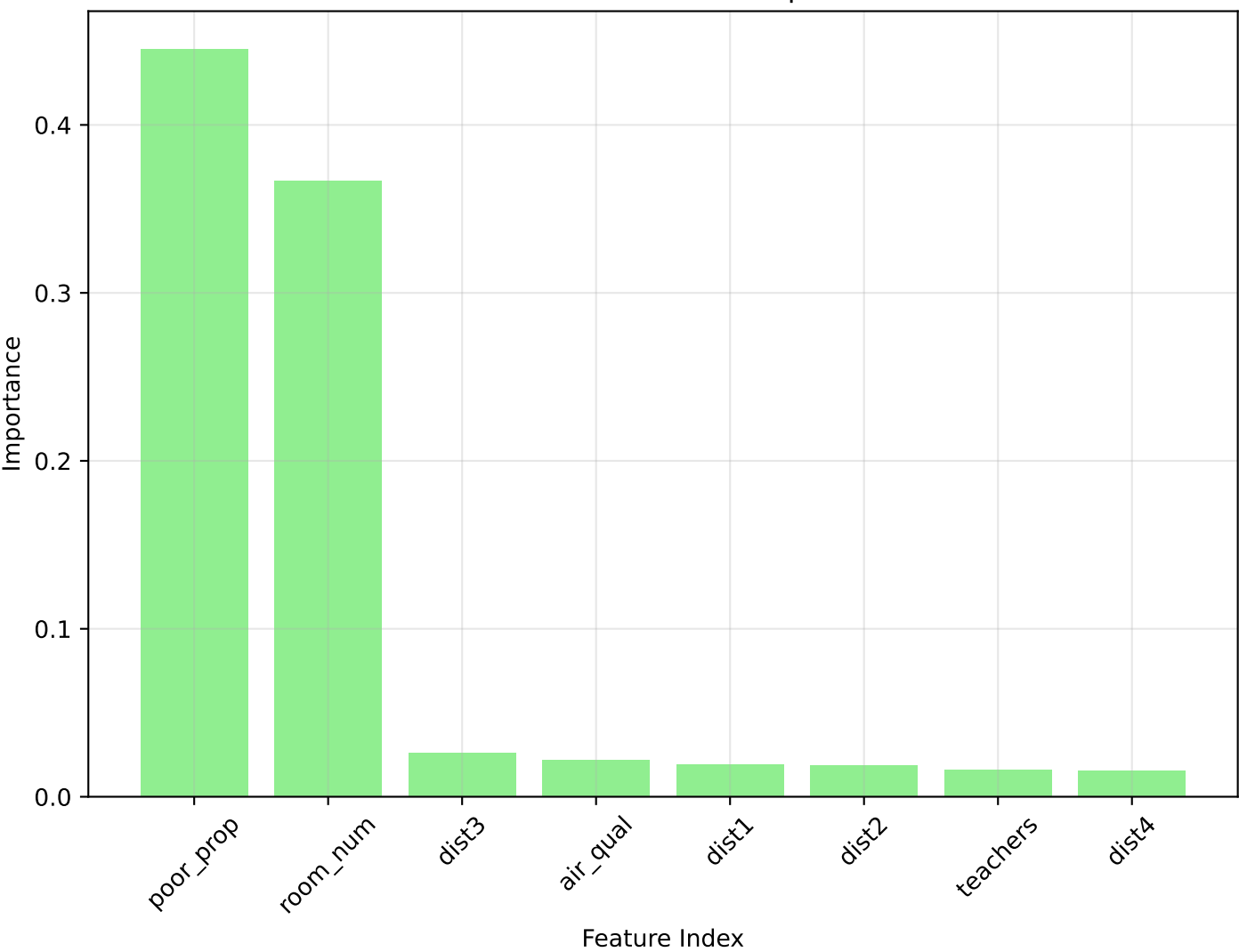
Model Performance Comparison



Best Model: Random Forest
($R^2 = 0.982$)



Random Forest Feature Importance



Residual Distribution Comparison

